Biologically meaningful between-sample comparison

Groups

Reference: WT. Comparison: KJ. Specimens: 1 and 1.

What to use

Descriptive comparison only. Inferential statistics are unavailable until each group has at least 3 biological specimens.

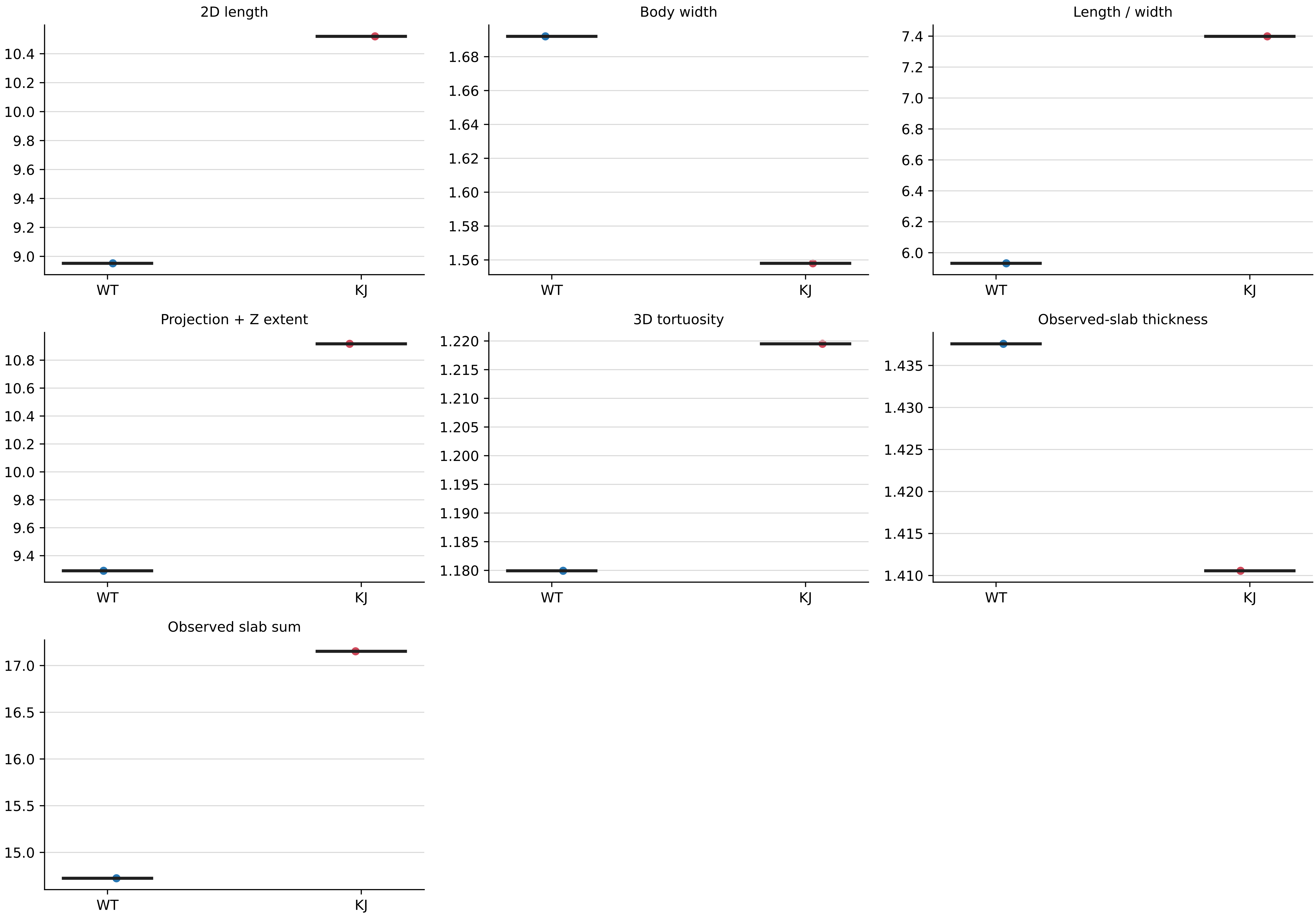
Scope

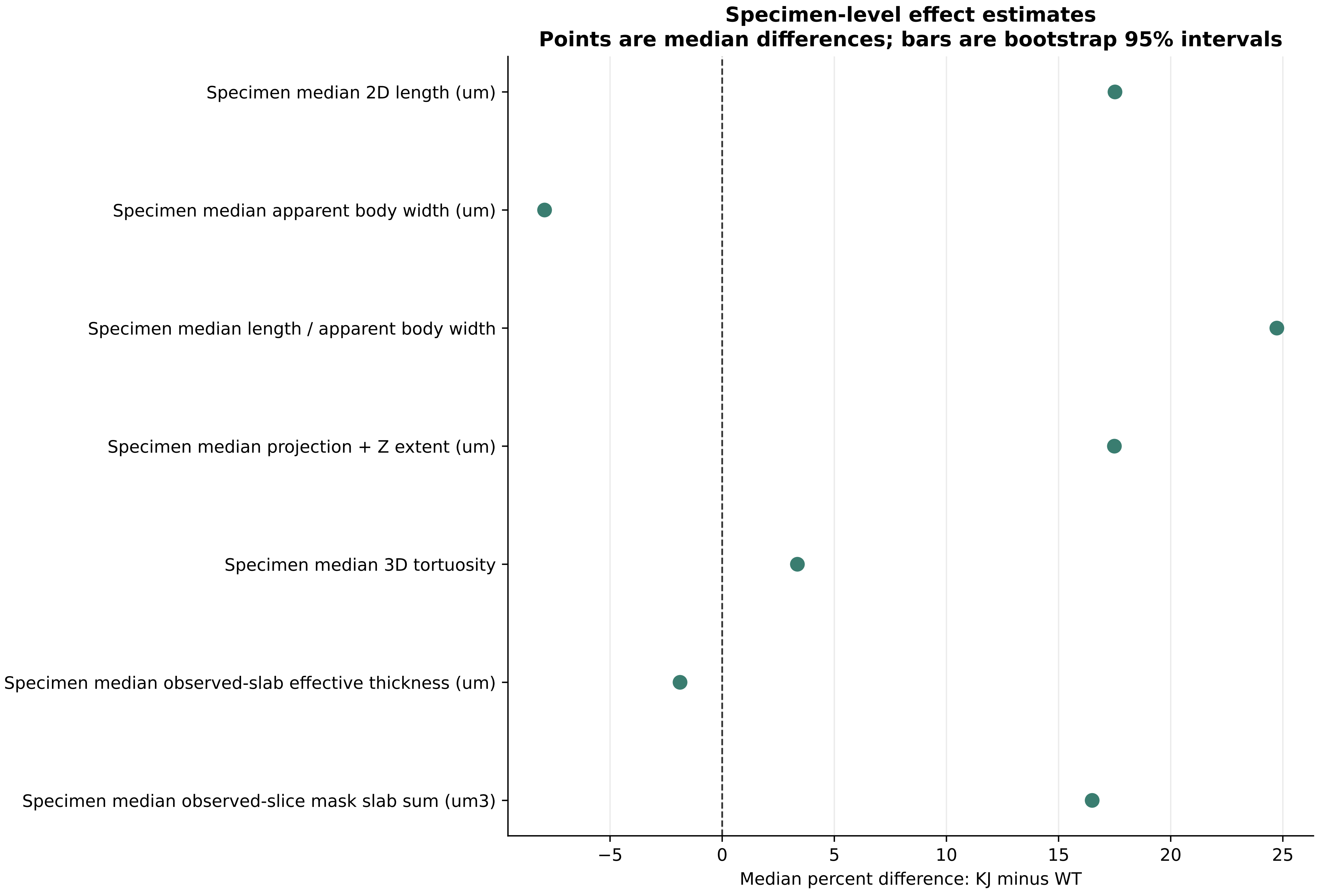
This report contains biological morphology comparisons only. Count, acquisition-depth, tracking, warning-category, and audit diagnostics are in the separate quality-control package.

Supporting files

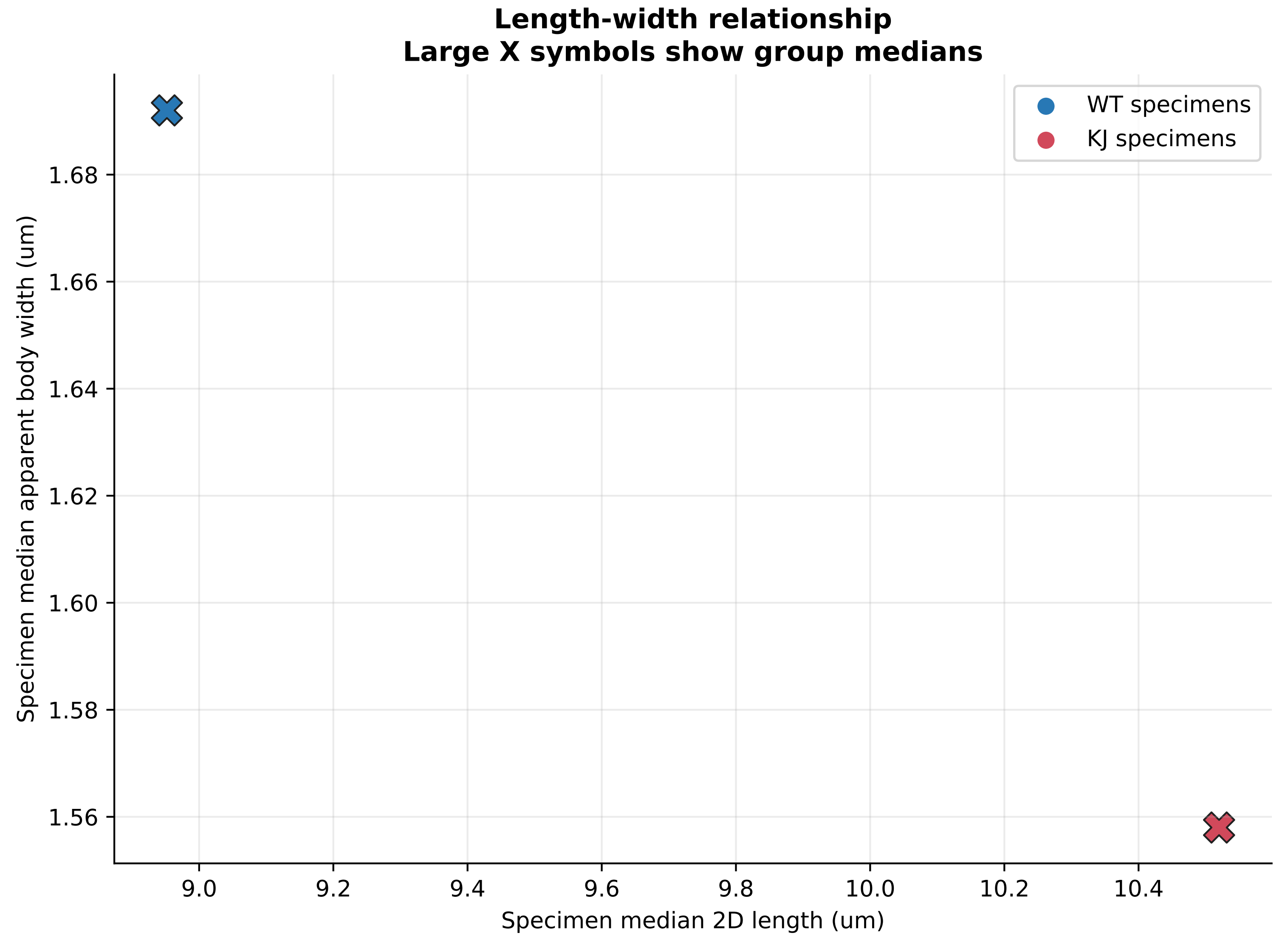
The accompanying Excel workbook contains every specimen-level value and every derived table plotted in this report.

Specimen-level biological measurements
Each point is one specimen; black bars show medians





Specimen-level effect estimates
Points are median differences; bars are bootstrap 95% intervals
Specimen median 2D length (um)
Specimen median apparent body width (um)
Specimen median length / apparent body width
Specimen median projection + Z extent (um)
Specimen median 3D tortuosity
Specimen median observed-slab effective thickness (um)
Specimen median observed-slice mask slab sum (um3)
−5
0
5
10
15
20
25
Median percent difference: KJ minus WT


Length-width relationship
Large X symbols show group medians
WT specimens
KJ specimens
Specimen median apparent body width (um)
Specimen median 2D length (um)
1.68
1.66
1.64
1.62
1.60
1.58
1.56
9.0
9.2
9.4
9.6
9.8
10.0
10.2
10.4

Specimen-level statistical summary
Each value is calculated across biological specimens, not pooled nuclei

Measure	WT median	KJ median	Median change (%)	Effect size Cliff's delta	Median test FDR q	Rank test FDR q	Mean test FDR q
Specimen median 2D length (um)	8.95	10.5	+17.5%	+nan	nan	nan	nan
Specimen median apparent body width (um	1.69	1.56	-7.9%	+nan	nan	nan	nan
Specimen median length / apparent body wid	5.93	7.4	+24.7%	+nan	nan	nan	nan
Specimen median projection + Z extent (um	9.29	10.9	+17.5%	+nan	nan	nan	nan
Specimen median 3D tortuosity	1.18	1.22	+3.4%	+nan	nan	nan	nan
cimen median observed-slab effective thicknes	1.44	1.41	-1.9%	+nan	nan	nan	nan
ecimen median observed-slice mask slab sum	14.7	17.2	+16.5%	+nan	nan	nan	nan

Medians: the typical specimen value in each group. Median change: (KJ median - WT median) / WT median x 100; positive means higher in KJ.
Cliff's delta: direction and separation, from -1 to +1; positive means KJ specimens tend to have higher values. FDR q: p-value adjusted for testing multiple measurements; q < 0.05 is evidence of a group difference.
Median test: permutation comparison of specimen medians (primary). Rank test: Mann-Whitney comparison of specimen ordering. Mean test: Welch comparison of specimen means.

Statistical methods and interpretation

Inference availability

Descriptive comparison only. Inferential statistics are unavailable until each group has at least 3 biological specimens.

Analysis unit

Each point and test uses one biological specimen. Individual nuclei are nested measurements and are not counted as independent replicates.

How to read the summary table

The two group medians describe the typical specimen in each group. Median change is the comparison-minus-reference difference expressed as a percentage of the reference median. Cliff's delta describes direction and group separation from -1 to +1. Each FDR q-value is the corresponding test's p-value after correction for testing multiple measurements; q below 0.05 is the conventional evidence threshold.

Primary inference

The two-sided permutation test compares the difference in specimen medians. Bootstrap intervals describe uncertainty in that median difference.

Mann-Whitney U

A secondary rank-based sensitivity test. It asks whether values from one group tend to rank above values from the other; it is not strictly a test of medians unless distribution shapes are similar.

Welch's t-test and ANOVA

Welch's t-test is a secondary mean-based sensitivity test and does not require equal variances. With exactly two groups, ordinary one-way ANOVA adds no independent information because it is equivalent to the equal-variance two-sample t-test.

Multiple tests

Benjamini-Hochberg FDR q-values are reported separately for each test family. Effect sizes and confidence intervals should be interpreted alongside q-values.

Biological meaning of the primary measurements

Specimen median 2D length (um)

Are nuclei typically longer in the image plane? The specimen median of each reconstructed nucleus's maximum calibrated 2D centerline length. Higher values indicate longer projected nuclei.

Specimen median apparent body width (um)

Are nuclei typically broader or thinner? The specimen median of representative-plane subpixel perpendicular body chords. Higher values indicate broader apparent masks; width remains sensitive to PSF, focus, and mask boundaries.

Specimen median length / apparent body width

Are nuclei more elongated or more rounded? A shape ratio. Higher values indicate longer, more slender nuclei; interpret it together with length and width.

Specimen median projection + Z extent (um)

Do nuclei have a larger combined image-plane and Z extent? A calibrated hypotenuse combining maximum lateral centerline extent with Z span. It is orientation-sensitive and is not an integrated 3D centerline.

Specimen median 3D tortuosity

Are nuclei straighter or more curved? Estimated path length divided by end-to-end distance. Values near 1 are straight; larger values indicate increasing curvature.

Specimen median observed-slab effective thickness (um)

Is the observed mask slab effectively thicker? A diameter proxy derived from volume divided by length. It is not a direct width and is PSF- and segmentation-sensitive.

Specimen median observed-slice mask slab sum (um3)

How much calibrated observed-slice mask slab does a typical nucleus occupy? Filled-mask area on observed planes multiplied by Z spacing, with no missing-plane interpolation. It depends on thresholds, sampling, and PSF.

Numerical source contract

These exact values are derived from specimen-level rows and are also stored in the Excel workbook and report_numeric_contract.json.

```
SOURCE_VALUE median_2d_length_um REF=8.952 COMP=10.52
SOURCE_VALUE median_body_width_um REF=1.692 COMP=1.558
SOURCE_VALUE median_length_body_width_ratio REF=5.9315 COMP=7.39855
SOURCE_VALUE median_projection_z_extent_um REF=9.29183 COMP=10.9169
SOURCE_VALUE median_3d_tortuosity REF=1.17993 COMP=1.2195
SOURCE_VALUE median_observed_slab_effective_thickness_um REF=1.43757 COMP=1.41056
SOURCE_VALUE median_observed_slice_mask_volume_um3 REF=14.7234 COMP=17.1525
```